

A Five-Block Obstruction and a Totally Real Degree-Fourteen Trace Field for Hénon Galois Excess

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Abstract

For the area-preserving Hénon recurrence $q_{j+1} = 1 - 6q_j^2 - q_{j-1}$, a previously constructed pair of primitive symbolic families satisfies an exact cyclic block-incidence identity at every width. We resolve the next metric member of this ladder. Reversibility reduces the period-six orbit A_6 to a degree-six coordinate equation and the period-seven orbits A_7, B_7 to two real embeddings of one degree-fourteen equation. Chronological monodromy elimination gives an irreducible cubic trace polynomial for A_6 and an irreducible totally real degree-fourteen trace polynomial shared by A_7, B_7 . Sturm intervals and an elementary logarithmic comparison prove

$$\mathcal{E}(A_5) + \mathcal{E}(B_7) - \mathcal{E}(A_6) - \mathcal{E}(B_6) > 0,$$

with numerical value $139.73257\dots$. The sign proof reduces to a strict integer-product comparison, independent of floating-point root separation. Hence no locally constant potential depending on at most five consecutive states realizes all primitive-orbit Galois excesses. A determinant-one width-six incidence minor interpolates the cumulative finite witness, so the unrestricted Hölder question remains open. We state the precise asymptotic discrepancy theorem still needed and preserve the physical-subsystem/arithmetical claim boundary.

1 Introduction

Periodic-orbit weights in hyperbolic dynamics are often generated by Hölder observables, and locally constant observables provide the finite memory models on a symbolic presentation [1, 3]. This elementary fact creates a sharp consistency test: whenever cyclic block-incidence rows satisfy an integer relation, the corresponding periodic sums of every potential of that memory must satisfy the same relation.

The HCS-P54 project split the multiplier Mahler height on a certified mixing area-preserving Hénon survivor into a physical instability length and a nonnegative Galois excess $\mathcal{E}(\gamma)$ [4]. HCS-P55 found the first three-block violation, and HCS-P56 replaced isolated searches by the primitive families

$$A_m = 0^{m-2}21, \quad B_m = 0^{m-3}231, \tag{1}$$

which obey

$$\mathbf{N}_m(A_m) + \mathbf{N}_m(B_{m+2}) = \mathbf{N}_m(A_{m+1}) + \mathbf{N}_m(B_{m+1}) \quad (m \geq 3). \tag{2}$$

The width-four member excludes every memory-at-most-four realization [5, 6].

This paper resolves the next member. Its algebraic input is not a black-box orbit solve: reversibility converts the required six- and seven-cycles into one-dimensional elimination problems. The seven-cycle problem is especially useful because the two physical words A_7 and B_7 occur as different real embeddings of one irreducible trace field.

Our main conclusion is the following.

Theorem 1.1 (Five-block obstruction). *On the frozen $H6$ survivor, no locally constant potential depending on at most five consecutive symbolic states has primitive-periodic sums equal to the Galois excess $\mathcal{E}(\gamma)$ for every primitive orbit γ . For the four witnessing cycles,*

$$\Delta_5 := \mathcal{E}(A_5) + \mathcal{E}(B_7) - \mathcal{E}(A_6) - \mathcal{E}(B_6) = 139.7325728699720846 \dots > 0.$$

The sign is certified by exact polynomial root isolation and an integer comparison, not by the displayed decimal.

The theorem is deliberately finite-memory. A width-six unimodular minor fits all cumulative witness values through this stage. Finite periodic data can also be interpolated by a general Hölder function on disjoint periodic orbits. Thus Theorem 1.1 is not an unrestricted Hölder no-go and not a Hilbert–Pólya construction. Its value is to move the exact regularity frontier and to expose a tractable reflection-chain asymptotic problem.

2 Frozen survivor, trace, and excess

We use the area-preserving map

$$H_6(q, p) = (1 - 6q^2 - p, q), \tag{3}$$

restricted to the four-state mixing survivor certified in the preceding projects. A periodic orbit with first coordinates $(q_0, \dots, q_{n-1})$ satisfies

$$q_{j+1} = 1 - 6q_j^2 - q_{j-1}, \quad j \pmod{n}. \tag{4}$$

The chronological derivative product is assembled from

$$D(q) = \begin{pmatrix} -12q & -1 \\ 1 & 0 \end{pmatrix}, \quad \det D(q) = 1. \tag{5}$$

If T_γ denotes its trace, the two multiplier roots satisfy $z + z^{-1} = T_\gamma$.

For an irreducible trace polynomial with real embeddings $\theta_1, \dots, \theta_d$, all of absolute value greater than two in the cases below, put

$$L(\theta) = \operatorname{arcosh}\left(\frac{|\theta|}{2}\right). \tag{6}$$

The multiplier Mahler height is $\sum_i L(\theta_i)$. For a selected physical embedding θ_{phys} , the Galois excess is

$$\mathcal{E}(\gamma) = \sum_i L(\theta_i) - L(\theta_{\text{phys}}) \geq 0. \tag{7}$$

This is the P54 convention. We do not rescale it or fit any values.

For a word w , let $\mathbf{N}_r(w)$ be its cyclic incidence vector of length- r blocks. Pairing a width- r potential with $\mathbf{N}_r(w)$ gives its periodic sum. The P56 insertion theorem gives (2) for every $m \geq 3$; the present proof uses its $m = 5$ instance.

3 Two reflection-reduced closing equations

Set

$$b = \frac{1 - 6a^2}{2}, \quad c = 1 - 6b^2 - a, \quad d = 1 - 6c^2 - b. \quad (8)$$

These are successive uses of (4) starting at a reflection axis.

3.1 The six-cycle A_6

The vertex-vertex reflection pattern is

$$(q_0, \dots, q_5) = (a, b, c, d, c, b). \quad (9)$$

The remaining endpoint equation is $1 - 6d^2 - 2c = 0$. Its exact factorization contains three quadratic factors, one quartic factor, and the primitive factor

$$f_6(a) = 2916a^6 - 1782a^4 + 108a^3 + 279a^2 - 33a - 2. \quad (10)$$

Reduction of all six recurrence residuals modulo f_6 gives zero. The physical coordinate is the unique root in

$$\left(\frac{551907131}{10^9}, \frac{551907132}{10^9} \right),$$

on which the chain signs are $(+, -, -, -)$. The full cyclic orbit therefore has exactly one positive coordinate, as required by $A_6 = 000021$.

3.2 The shared seven-cycle chain

The edge-reflection pattern is

$$(q_0, \dots, q_6) = (a, b, c, d, d, c, b). \quad (11)$$

Now the last equation is $1 - 6d^2 - c - d = 0$, and

$$1 - 6d^2 - c - d = -\frac{1}{2}(6a^2 + 2a - 1)f_7(a), \quad (12)$$

where f_7 is the degree-fourteen polynomial in Appendix A. Every recurrence residual vanishes modulo f_7 .

Two rational boxes select the two physical embeddings:

$$\begin{aligned} B_7 : \quad a &\in \left(-\frac{600956965}{10^9}, -\frac{600956964}{10^9} \right), & \text{sgn}(a, b, c, d) &= (-, -, -, +), \\ A_7 : \quad a &\in \left(\frac{551935742}{10^9}, \frac{551935743}{10^9} \right), & \text{sgn}(a, b, c, d) &= (+, -, -, -). \end{aligned}$$

In the full palindromic patterns these are, up to cyclic rotation, the two-positive word B_7 and the one-positive word A_7 .

4 Cubic and degree-fourteen trace fields

Proposition 4.1 (The A_6 trace field). *Modulo $f_6(a)$, the chronological trace is*

$$T_6(a) = -2(1364688a^4 - 94608a^3 - 379080a^2 + 56808a + 689).$$

Its resultant with f_6 is the square of

$$F_6(T) = T^3 + 48342T^2 - 334511988T + 306994257352. \quad (13)$$

The polynomial F_6 is irreducible over $\mathbb{Q}$, has three real roots, one in each interval

$$(-54575, -54574), \quad (1094, 1095), \quad (5137, 5138), \quad (14)$$

and the first is the physical trace of A_6 .

Proof. Direct matrix multiplication followed by polynomial remainder gives the displayed trace. Elimination gives F_6^2 ; the square corresponds to the two choices of reflection-axis coordinate for each orbit. Reduction modulo 5 is irreducible. Exact Sturm counts give one root in each interval in (14), exhausting the degree. The rational coordinate box from Section 3 maps strictly inside the first trace interval. $\square$

The reciprocal multiplier polynomial is

$$z^6 + 48342z^5 - 334511985z^4 + 306994354036z^3 - 334511985z^2 + 48342z + 1, \quad (15)$$

and is irreducible over $\mathbb{Q}$.

Proposition 4.2 (The shared A_7/B_7 field). *Eliminating a from the period-seven trace and $f_7(a)$ gives an irreducible degree-fourteen polynomial $F_7(T)$. It has fourteen real roots, isolated by the intervals in Appendix A. The physical embeddings satisfy*

$$T(A_7) \in (-390512, -390511), \quad T(B_7) \in (230985, 230986).$$

Thus A_7 and B_7 are physical real embeddings of one totally real degree-fourteen trace field. Its reciprocal multiplier polynomial is irreducible of degree twenty eight.

Proof. The exact reduced trace is printed in the certificate. Its resultant is the polynomial F_7 in Appendix A; reduction modulo 37 is irreducible. Sturm counts give one root in each of fourteen disjoint integer intervals. The two coordinate boxes of Section 3 map inside the claimed trace intervals and carry the required sign words. Substitution $T = z + z^{-1}$, followed by multiplication by z^{14} , gives the reciprocal degree-twenty-eight polynomial; exact factorization over $\mathbb{Q}$ is irreducible. $\square$

Numerically, for orientation only,

$$\begin{aligned} \mathcal{E}(A_6) &= 15.54277149078026246\dots, \\ \mathcal{E}(A_7) &= 129.0589652116398587\dots, \\ \mathcal{E}(B_7) &= 129.5840670640364218\dots \end{aligned}$$

No theorem below depends on rounding these values.

5 An exact five-block obstruction

We first turn coarse trace isolators into a rigorous separation.

Lemma 5.1 (Elementary logarithmic bracket). *For every real $u > 2$,*

$$\log(u - 1) < \operatorname{arcosh}(u/2) < \log u.$$

Proof. The expanding root of $z + z^{-1} = u$ is $\lambda = (u + \sqrt{u^2 - 4})/2$. Since $u - 2 < \sqrt{u^2 - 4} < u$, one has $u - 1 < \lambda < u$. Taking logarithms proves the claim. $\square$

For A_5 , the physical trace is the root in $(-7607, -7606)$. The other five locked intervals are

$$(-711, -710), (-590, -589), (390, 391), (770, 771), (4445, 4446).$$

Lemma 5.1 gives

$$\mathcal{E}(A_5) > \log(709 \cdot 588 \cdot 389 \cdot 769 \cdot 4444). \quad (16)$$

For A_6 , Proposition 4.1 gives

$$\mathcal{E}(A_6) < \log(1095 \cdot 5138). \quad (17)$$

P56 proved $\mathcal{E}(B_6) = \operatorname{arcosh}(9031 - 2676\sqrt{7})$ and $1950 < 9031 - 2676\sqrt{7} < 1951$. Hence

$$\mathcal{E}(B_6) < \log 3902. \quad (18)$$

The required product comparison is exactly

$$709 \cdot 588 \cdot 389 \cdot 769 \cdot 4444 = 554208972546768, \quad (19)$$

$$1095 \cdot 5138 \cdot 3902 = 21953081220, \quad (20)$$

$$554208972546768 - 21953081220 = 554187019465548 > 0. \quad (21)$$

It follows that

$$\mathcal{E}(A_5) > \mathcal{E}(A_6) + \mathcal{E}(B_6). \quad (22)$$

Moreover $\mathcal{E}(B_7) > 0$, since the irreducible degree-fourteen field has nonphysical expanding trace embeddings. Thus

$$\Delta_5 = \mathcal{E}(A_5) + \mathcal{E}(B_7) - \mathcal{E}(A_6) - \mathcal{E}(B_6) > 0. \quad (23)$$

Proof of Theorem 1.1. The $m = 5$ member of (2) is

$$\mathbf{N}_5(A_5) + \mathbf{N}_5(B_7) = \mathbf{N}_5(A_6) + \mathbf{N}_5(B_6).$$

Every width-five potential must therefore give $\Delta_5 = 0$, contrary to (23). A potential of any shorter width is also a width-five potential after ignoring the extra coordinates. This excludes all widths one through five. $\square$

The high-precision diagnostic value is

$$\Delta_5 = 139.7325728699720846164890241354 \dots,$$

but its only logical role is reproducibility; equations (16)–(21) certify the sign.

6 Finite sharpness at width six

An obstruction from four rows can be an artifact of using too-short cylinders. Here that limitation is exact.

Proposition 6.1 (Four-row sharpness). *For row order (A_5, B_7, A_6, B_6) , the width-five incidence matrix has rank three and left-kernel vector $(1, 1, -1, -1)$. At width six, the columns*

$$000021, \quad 000023, \quad 000210, \quad 000231$$

form the minor

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \det = -1.$$

Consequently arbitrary totals on these four cycles are realizable by a width-six cylinder function.

The finite sharpness persists after including all earlier obstruction rows.

Proposition 6.2 (Cumulative sharpness). *For the nine rows*

$$C_1, A_3, A_4, B_4, A_5, B_5, B_6, A_6, B_7,$$

there is a selected nine-column width-six incidence minor of determinant one. Hence every assignment of nine periodic totals, including their exact Galois excesses, has a unique interpolation on those selected cylinders.

Proof. The selected columns and full integer matrix are retained in the primary certificate. Fraction-free elimination gives determinant one. An independent implementation rebuilds the block counters and checks the four-row determinant without importing the primary code. $\square$

Propositions 6.1 and 6.2 prevent a common overclaim. Theorem 1.1 raises the minimum possible local memory to six; it does not prove that width six works on every primitive orbit.

7 The remaining Hölder gate

For one-sided symbolic metric $d_{\vartheta}(x, y) = \vartheta^k$, the P56 ladder pairs orbits whose long zero runs agree except near a bounded number of seams. If one Hölder observable of exponent $\alpha > 0$ realized all excesses, its variation estimate would force

$$|\Delta_m| \leq C(4m + 4)\vartheta^{\alpha m} \quad (m \geq 3) \tag{24}$$

for one fixed $C > 0$.

The new value $\Delta_5 \neq 0$ is not a contradiction to (24): a constant C can absorb any finite number of indices. The asymptotic gate is therefore precise.

Determine the leading stable-tail mode of the two reflection chains and show either that Δ_m obeys a compatible exponential law or that an infinite subsequence violates every bound of the form (24).

The recurrence has a negative fixed-point tail, so the natural next attack is to linearize the closing equations about that fixed point and compare the two reflection boundary conditions. This is a whole-family theorem, not a request for another isolated period. Computing B_8 is useful only as a falsifier for the predicted leading eigenmode.

For a two-sided Hölder observable, a future-dependent representative would first require a cohomological reduction. We make no such reduction here. Livšic theory tests equality modulo coboundaries between already-defined Hölder observables; it does not construct an observable from arbitrary periodic totals [2].

8 Reproducibility and route evaluation

The primary certificate performs the following exact checks:

1. the primitive cycle census through period seven and the all-width symbolic ladder through width 64 as a finite regression;
2. both reflection closing factorizations and every recurrence residual;
3. chronological trace reduction and both resultants;
4. irreducibility witnesses modulo 5 and 37;
5. all rational coordinate and trace isolators;
6. the exact integer obstruction margin and high-precision diagnostic;
7. the width-five relation and two width-six unimodular minors;
8. eight dependency hashes and twenty-two adversarial mutations.

An independent script derives the factors and resultants afresh. Fifteen unit tests lock the public result schema and claim boundary. All commands are collected in `code/run.c57.sh`.

Under the frozen Route-A evaluator, the status remains

$$(A1_{\text{weak}}, A2_{\text{physical subsystem}}, A3_{\text{partial}}, A4_{\text{formal hint}}),$$

with overall verdict `ROUTE_A_EXPLORATORY`. The physical suspension zeta inherited from earlier work remains the only analytic determinant. The Galois-excess completion has no all-period Hölder observable and therefore no full A2 pass.

Route B is not authorized. No Hilbert space, self-adjoint operator, von-Mangoldt trace formula, completed determinant or divisor equality is constructed. The exact trace fields are algebraic dynamical data, not a rational-prime correspondence.

A Exact polynomial and isolation ledger

The period-seven coordinate factor is

$$\begin{aligned} f_7(a) = & 612220032a^{14} - 204073344a^{13} - 646232256a^{12} + 226748160a^{11} \\ & + 247533408a^{10} - 90069408a^9 - 41045616a^8 + 15046560a^7 \\ & + 2834352a^6 - 948672a^5 - 97848a^4 + 22032a^3 + 1512a^2 - 144a - 7. \end{aligned}$$

The trace polynomial is

$$\begin{aligned}
F_7(T) = & T^{14} + 53380T^{13} - 116230063064T^{12} + 8748136108873472T^{11} \\
& + 930148180553911001792T^{10} - 55441728431344376174558464T^9 \\
& - 1042857805960956394737247010304T^8 \\
& + 67800369626486919391549995220844544T^7 \\
& + 248063550855956838882026032464056057856T^6 \\
& - 18599810233678146836278775387225818293010432T^5 \\
& - 125626107241488921782218909595028406082053341184T^4 \\
& + 648460503108807187082799707783053721181729242218496T^3 \\
& + 5188095852189806429800801204477225625428281884227403776T^2 \\
& - 3780455159472389946318383487211963920344828975222028763136T \\
& - 43775419253348240272917680080620537154762693674649008390275072.
\end{aligned}$$

Its fourteen Sturm isolators, in increasing order, are

$$\begin{aligned}
& (-390512, -390511), (-76494, -76493), (-33929, -33928), \\
& (-9534, -9533), (-9431, -9430), (-5707, -5706), (-4082, -4081), \\
& (3217, 3218), (5681, 5682), (29838, 29839), (32741, 32742), \\
& (36376, 36377), (137464, 137465), (230985, 230986).
\end{aligned}$$

Each interval contains exactly one root; their count exhausts the degree.

The numerical excess ledger is

orbit	Galois excess
A_5	33.9605061147770865686214017363...
A_6	15.5427714907802624626376595709...
B_6	8.2692288180611612460532607668...
B_7	129.584067064036421756558542737...
Δ_5	139.732572869972084616489024135...

Only the exact isolators and integer margin, not these decimals, prove the obstruction.

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